Steady-State Heat Transfer in a Rectangular Plate: Jacobi and Point Gauss-Seidel Methods

1. Problem statement

A rectangular plate of size $L_x = 2$ m × $L_y = 1$ m is held at $T = 1000$ K on the left wall and $T = 500$ K on the right wall. The top and bottom walls are adiabatic. Find the steady-state temperature distribution on a uniform grid with $\Delta x = \Delta y = 0.01$ m, plot the temperature contours, and plot the temperature along x at $y = 0.5$ m. Solve with (a) the Jacobi method and (b) the Point Gauss-Seidel (PGS) method, and compare the number of iterations required for an accuracy of 0.1 K.

2. Governing equation and discretisation

Steady conduction with constant properties and no source is governed by the Laplace equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0.$$

With $\Delta x = \Delta y$, the second-order central-difference form is

$$T_{i,j} = \tfrac{1}{4}(T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}).$$

Boundary conditions

- Left/right (Dirichlet): $T_{0,j} = 1000$ K, $T_{m,j} = 500$ K.
- Top/bottom (Neumann, adiabatic): $\partial T/\partial y = 0$. A ghost row is placed outside each wall and set by a central difference, $\dfrac{T_{i,,\text{above}} - T_{i,,\text{below}}}{2\Delta y} = 0$, i.e. the ghost value equals the interior node one row beyond the wall (mirror condition).

3. Algorithm

Step 1 – Domain and grid:
$L_x = 2$ m, $L_y = 1$ m, $\Delta x = \Delta y = 0.01$ m, so $m = 200$ cells in x and $n = 100$ cells in y. This gives $(m+1) \times (n+1) = 201 \times 101$ physical nodes. Two ghost rows are added, so the array has $n + 3$ rows. Physical nodes are $j = 1, \ldots, n+1$ (so $y = (j-1)\Delta y$, and $y = 0.5$ m is row $j = 51$); ghost rows are $j = 0$ and $j = n + 2$.

Step 2 – Dirichlet boundaries: $T_{0,j} = 1000$ K (left), $T_{m,j} = 500$ K (right).

Step 3 – Initial guess: All other nodes are set to 750 K, the mean of the two wall temperatures.

Step 4 – Adiabatic boundaries (ghost rows):
At the start of every sweep, mirror the interior values of the previous iterate into the ghost rows:

- bottom wall ($j = 1$): $T_{i,0} = T_{i,2}$
- top wall ($j = n + 1$): $T_{i,n+2} = T_{i,n}$

Step 5 – Iterative update (sweep over $j = 1, \ldots, n + 1$ and $i = 1, \ldots, m - 1$):

- **Method A – Jacobi:** every node uses only values from the previous iteration:

$$T_{i,j}^{new} = 0.25 \left(T_{i-1,j}^{old} + T_{i+1,j}^{old} + T_{i,j-1}^{old} + T_{i,j+1}^{old} \right)$$

- **Method B – Point Gauss-Seidel:** nodes already updated in the current sweep are used immediately (sweep order: j outer loop, i inner loop):

$$T_{i,j}^{new} = 0.25 \left(T_{i-1,j}^{new} + T_{i+1,j}^{old} + T_{i,j-1}^{new} + T_{i,j+1}^{old} \right)$$

Step 6 – Convergence check:
Compute the RMS change between successive iterates

$$e = \sqrt{\frac{\sum \left(T_{i,j}^{new} - T_{i,j}^{old} \right)^2}{(m+1)(n+1)}}.$$

If $e > 10^{-5}$, swap the old/new arrays and return to Step 4. Otherwise stop, write the temperature field, and extract the row $j = 51$ ($y = 0.5$ m).

Choice of tolerance. The RMS *change* per iteration is not the same as the *error* in T, and for slowly converging iterations the error is much larger than the change. A tolerance of 10^{-5} was chosen because it gives a true error close to the required 0.1 K.

4. Test results

4.1 Iteration comparison (tolerance $e \leq 10^{-5}$)

Method	Total iterations
Jacobi	32195
Point Gauss-Seidel	22973

PGS required about 29 % fewer iterations than Jacobi (22973 / 32195 ≈ 0.71).

4.2 Temperature contours

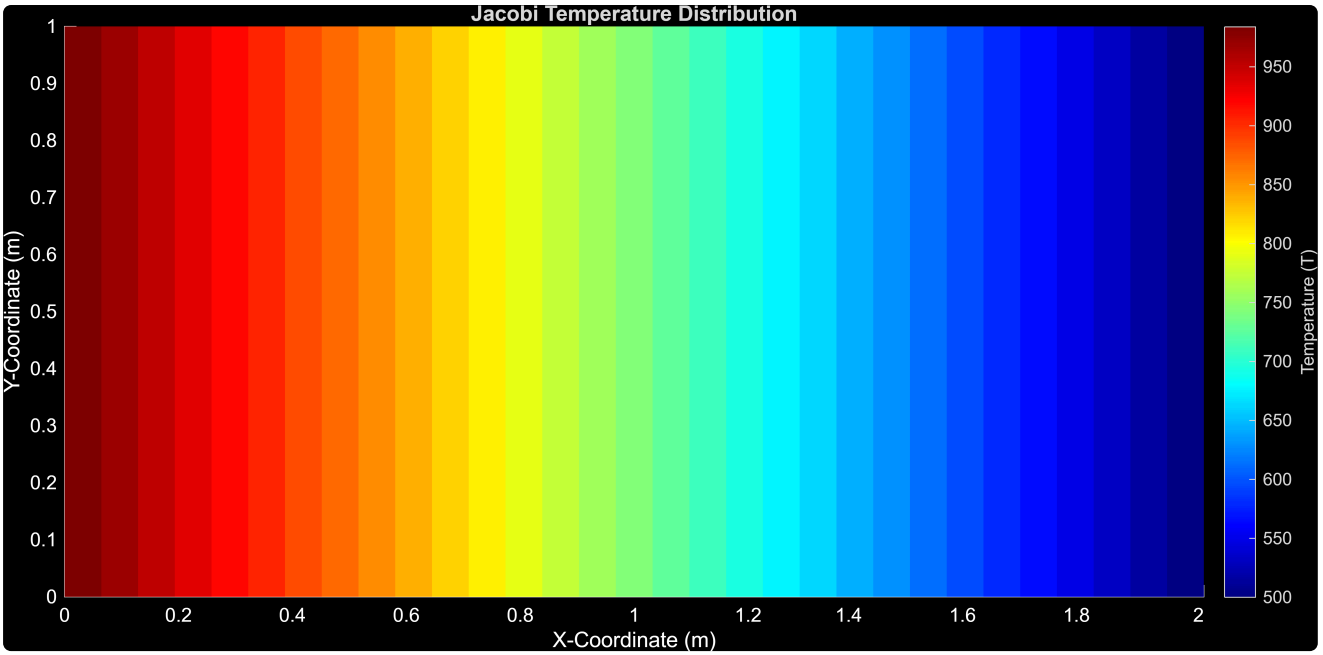


Figure 1: Steady-state temperature distribution (Jacobi).

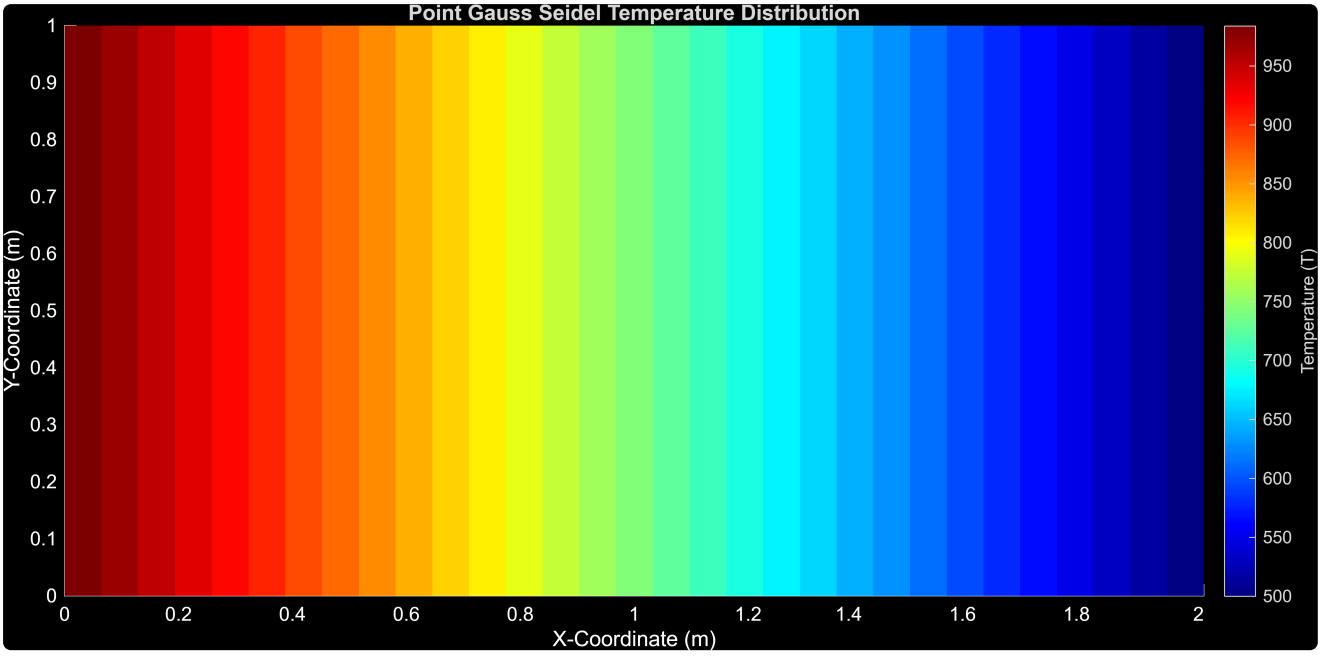


Figure 2: Steady-state temperature distribution (Point Gauss-Seidel).

The isotherms are vertical lines, i.e. T varies only with x. This is expected: the adiabatic top and bottom walls allow no heat flow in the y direction.

4.4 Centerline temperature distribution ($y = 0.5$ m)

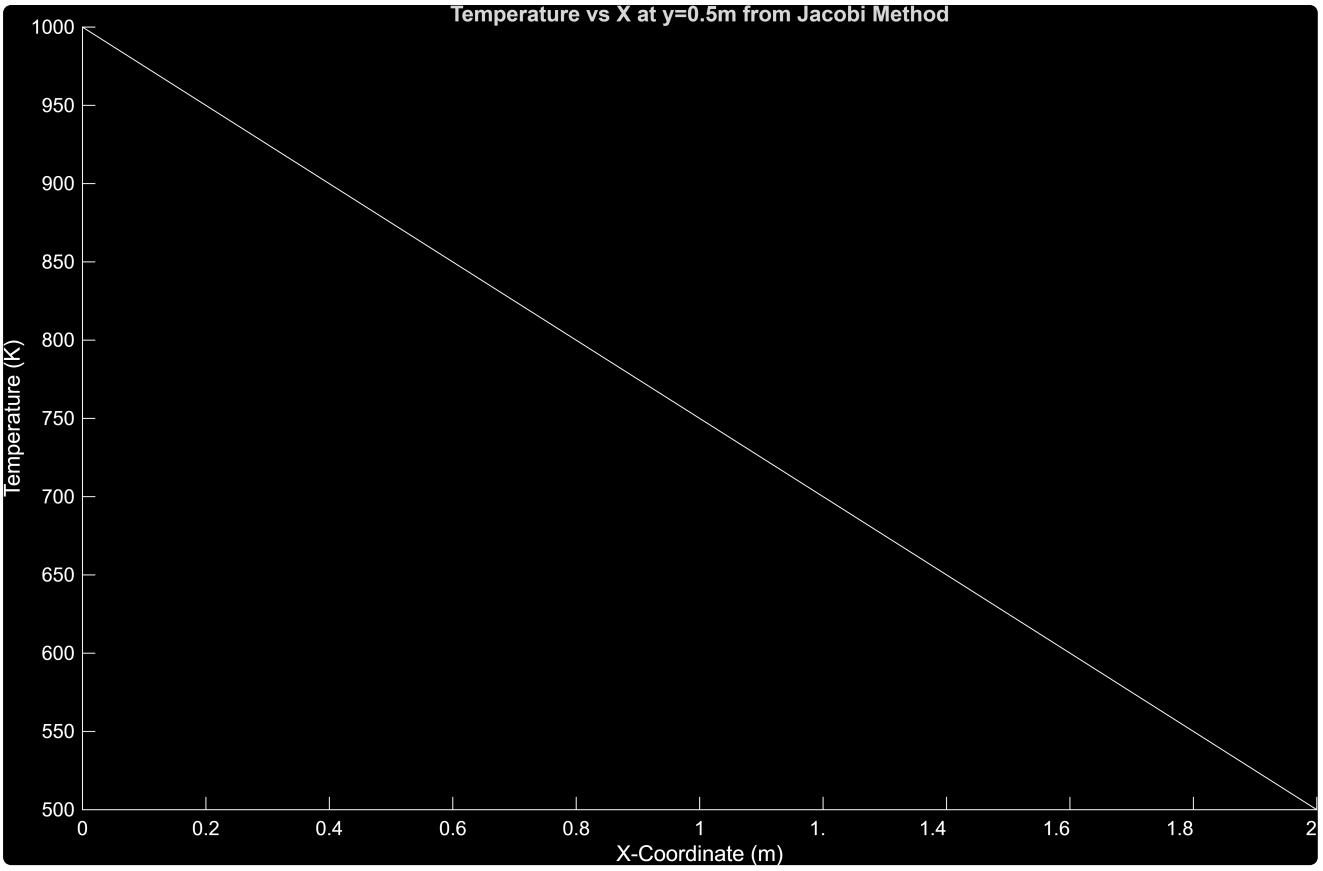


Figure 3: Temperature along x at $y = 0.5$ m (Jacobi).

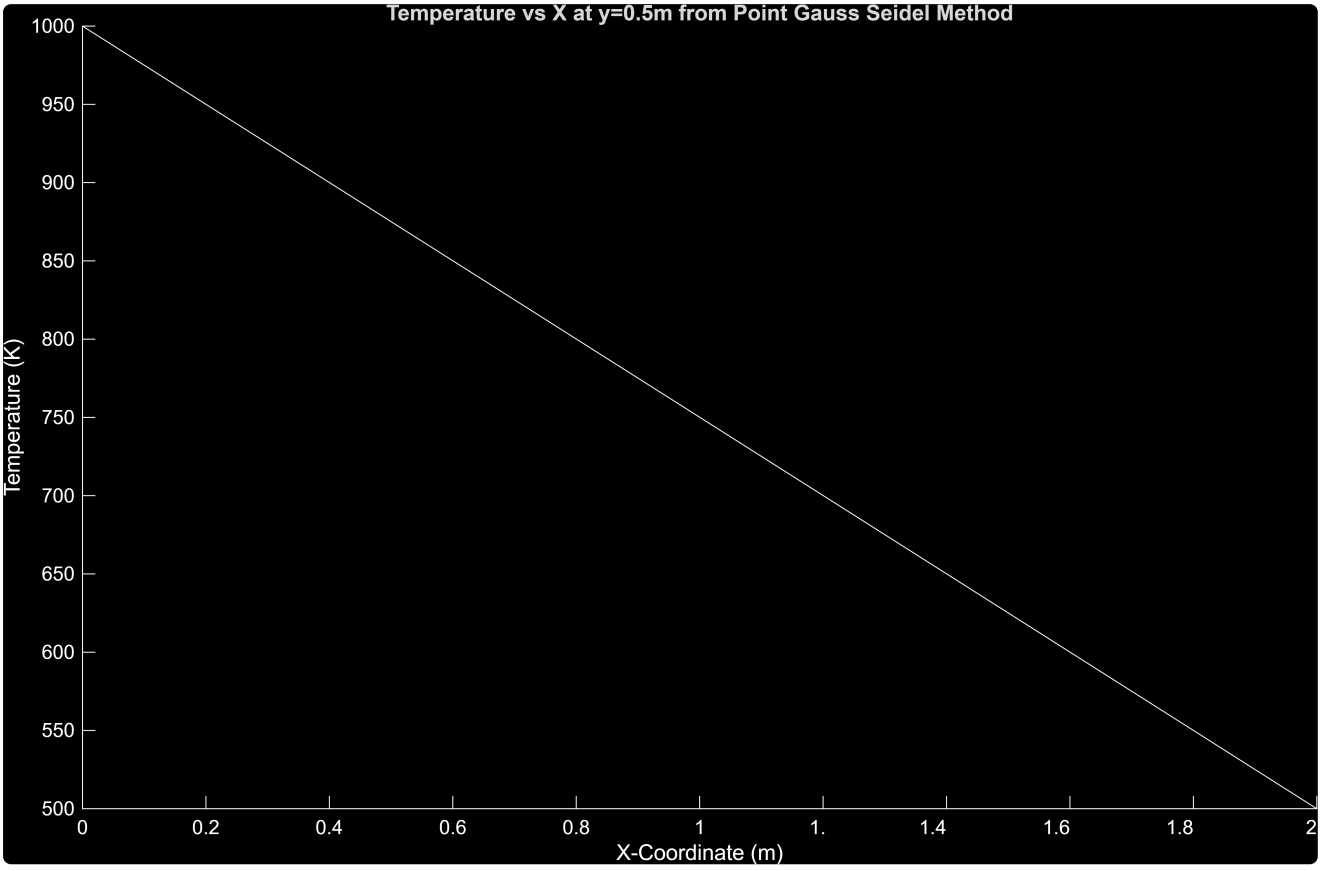


Figure 4: Temperature along x at $y = 0.5$ m (Point Gauss-Seidel).

The temperature falls linearly from 1000 K at $x = 0$ to 500 K at $x = 2$ m (gradient −250 K/m), matching the exact solution.

5. Discussion and conclusion

- Both methods converge to the same, essentially exact, linear temperature profile.
- Because PGS uses freshly updated neighbours, it needed fewer iterations than Jacobi (22973 vs 32195 at $e \leq 10^{-5}$).
- The RMS change per iteration underestimates the true error; the exact solution was therefore used to confirm that the final accuracy is about 0.1 K.

Code: `assignment_5.cpp` (code has been compiled with `g++ -O2 -o assignment_5.exe assignment_5.cpp`; it writes `jacobi.dat`, `pgs.dat`, `y_const_jacobi.dat`, `y_const_pgs.dat` to the current path).